

IN3160 Oblig 3

- a. The output is updated after 400ns, 200ns after indata is set.
The delay is caused by the intermediate signals, b and c, being updated after each next rising edge on mclk, since the value of a signal changes when the process has ended. These signals are set by variables v1 and v2, which update immediately in a process.
- b.
 - 1. outdata is now updated after 700ns and 800ns, with a 400ns delay after each indata update.
 - 2. The output delay doubled because I turned v1 and v2 into signals, which adds two extra 100ns delays.
 - 3. Since reset is turned off when starting the simulation and clk starts high, instead of rising, every signal and outdata is not set at start, and are therefore latched.
- c.
 - 1. output(7 downto 6) and output(3 downto 2) are both decided by signals sig1 and sig2, which are in turn decided by variables var1 and var2. Since a signal's value is set after a process has ended, it will be given the latest value of a variable, so sig1 is ultimately set to indata and sig2 to not indata, so that output(3 downto 2) and output(7 downto 6) will have the same values ("01" or "10").
 - 2. output(5 downto 4) and output(1 downto 0) are both decided by variables var1 and var2, which update immediately in a process. After output(1 downto 0) is set, var1 and var2 are inverted and output(5 downto 4) is set to those, which always makes these values different from each other ("00" and "11").
- d. Since sig1 and sig2 are removed from the sensitivity list, the output signal bits determined by these signals (7 downto 6, 3 downto 2) will not update immediately, resulting in the output signal bits being latched initially. They are not updated until when the input signal is changed, and be delayed compared to the rest of the output bits.

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